

Project Documentation

Project Title: Auto Ticket Classification using Flow Designer

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1.Introduction

Problem Statement

The school IT helpdesk receives multiple incident requests daily from students and teachers, such as Wi-Fi issues, projector failures, password problems, and slow computers.

Currently, IT staff manually reviews each request and assigns a category, which is time-consuming and inefficient.

This project automates ticket classification using ServiceNow Flow Designer by analysing keywords in the incident's short description and description fields.

Objective

- Automatically classify incidents at the time of creation.
- Reduce manual effort for IT agents.
- Improve ticket routing efficiency.
- Implement a no-code, easily maintainable solution.

PHASE 1: Requirement Analysis & Planning

The school IT helpdesk receives multiple support requests daily from students and teachers related to common issues such as network connectivity, hardware failures, account access problems, and system performance issues.

Currently, IT staff manually review each ticket to:

- Identify the issue type.
- Select the correct category and subcategory.
- Inform the caller about ticket creation.

This manual process is time-consuming, error-prone, and not scalable as the number of requests increases.

Activity : 1 Business Use Case.

Business Requirements

The system must:

1. Automatically classify IT tickets based on the issue description.
2. Assign both Category and Subcategory without manual intervention.
3. Support dependent choice logic between Category and Subcategory.
4. Send an automated email notification to the caller upon ticket creation.
5. Store ticket information in a structured and standardized format.
6. Ensure easy maintenance and future scalability.

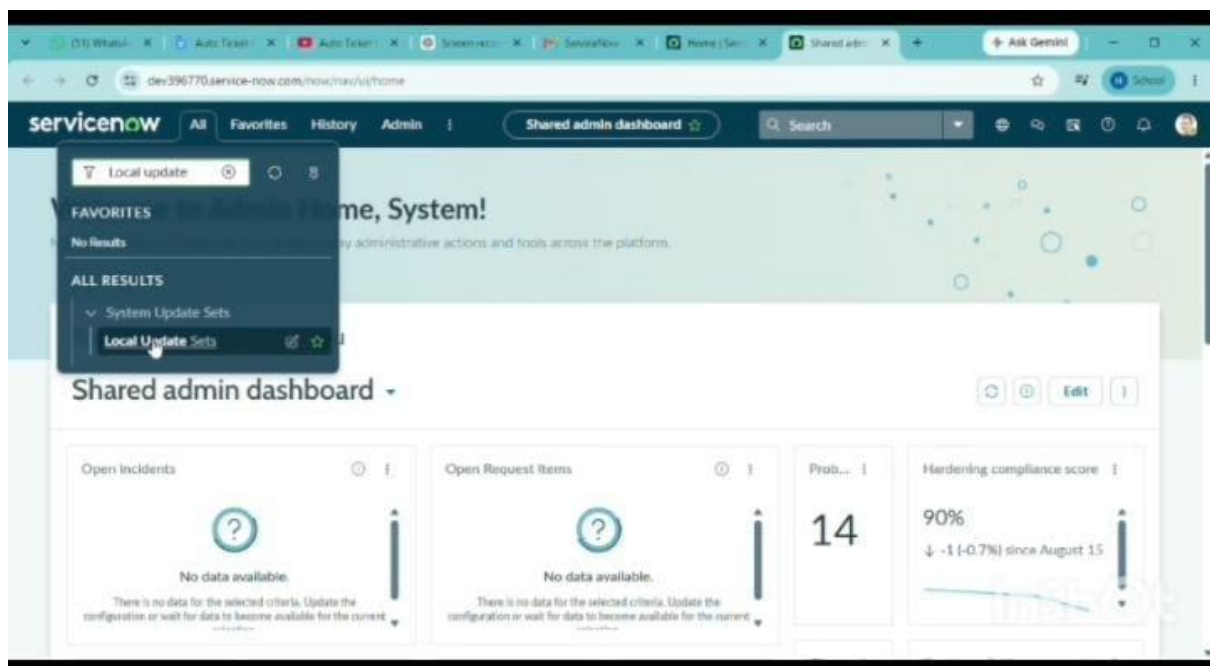
Activity : 2 Creation of New Update Set.

Navigation Steps :

- Go to All menus → Search for Update set.
- Under System Update Sets → Select Local Update Sets.
- Click the New button on the top right corner of the page.

- Name : **Project Update Set.**
- State : **In progress.**
- Application : **Global.**
- Then **Save** the **Form.**
- Under Related links Click on the **Make This My Current Set.**
- Then the Created Update Set will appear as Current.
-

Activity : 3 Navigation steps in Picture View.



Auto Ticket Classification using F... ServiceNow Developers ServiceNow

https://dev448132.service-now.com/now/nav/ui/classic/params/target/sys_update_set_list.do%3Fsysparm_userpref_module%3D50047c06da8016cd135a14cebc8...

servicenow All Favorites History Workspaces ServiceNow Search

Update Set
New record

Submit Submit and Make Current

* Name Project Update Set Application Global

State In progress

Parent

Release date

Description

Submit Submit and Make Current

(3) WhatsApp Auto Ticket Auto Ticket Screen rec... ServiceNow Home | Serv... Project Upc... Ask Gemini

dev396770.service-now.com/now/nav/ui/classic/params/target/sys_update_set.do%3Fsys_id%3Dc3f05714a03a7031043b6a43b0eeef3ca%2Fsysparm_view%3D...

servicenow All Favorites History Admin Update Set - Project Update Set Search

Update Set
Project Update Set

Your current update set has been changed to Default [Global]

Update Delete

* Name Project Update Set Application Global

State In progress

Parent

Release date

Install date

Installed from

Description

Created 2025-09-27 02:13:54

Created by admin

Merged to

Update Delete

Related Links

[Make This My Current Set](#)

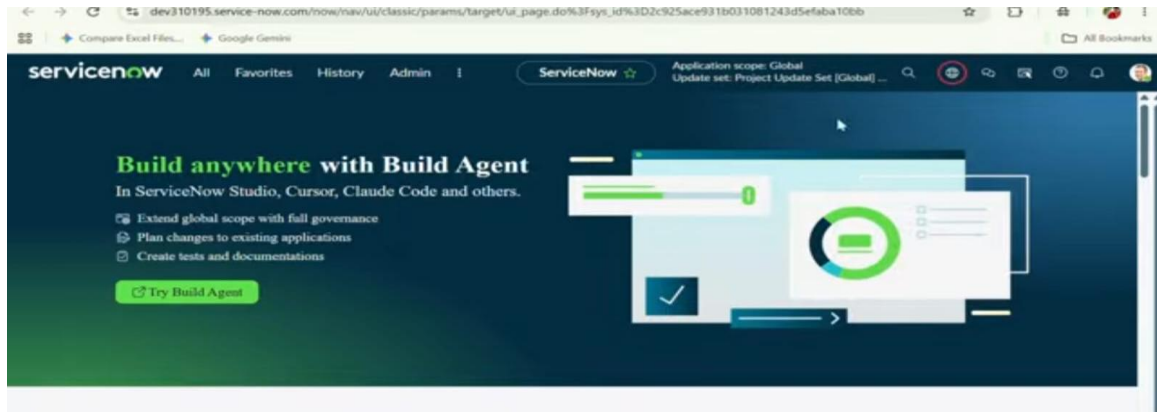
[Merge With Another Update Set](#)

[Scan Update Set](#)

Customer Updates Update Set Logs Child Update Sets Install History

Created Search

InShot

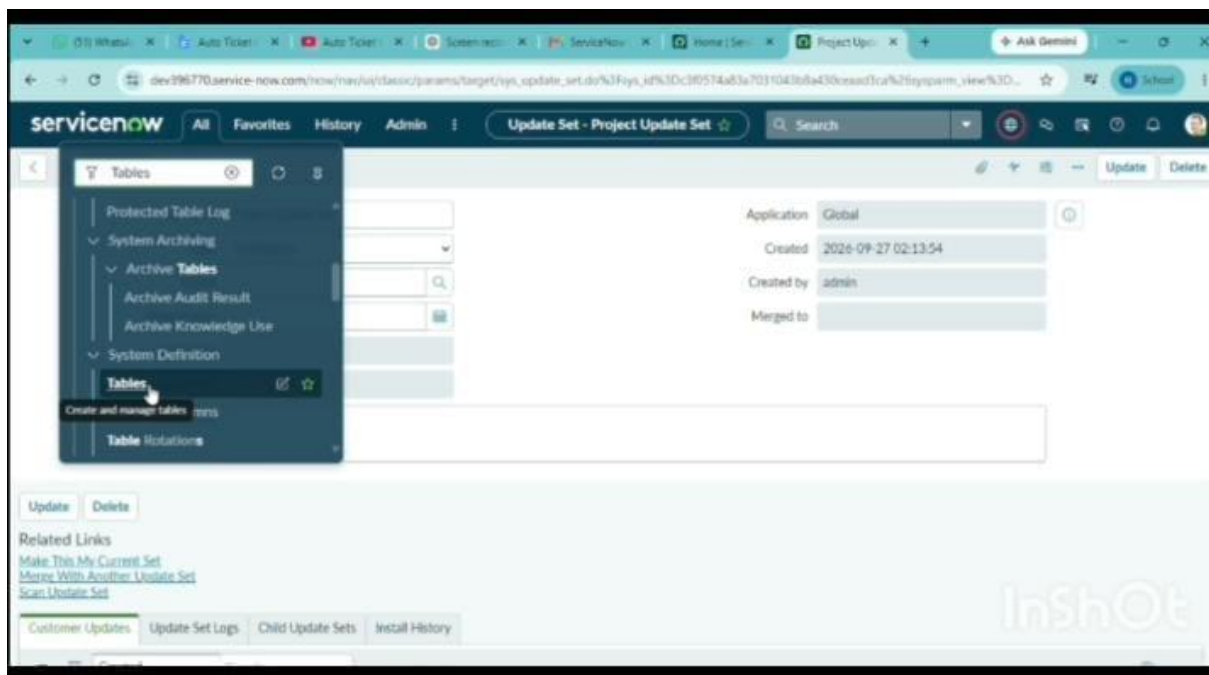


Phase :- 2 Backend Development & Configuration

Activity 1 : Creation of Custom Table to Store Ticket Records.

Steps to Create Custom Table

- Open All menus → Search for Tables.
- Under **System Definition** → Click **Tables**.
- On the top right corner Click **New** button.
- Label : **Incident WorkFlow**.
- Uncheck the **Create module**.
- Now **Save** the Form.
- Then Click the **Controls** Suction and Enable the Check box **true Auto Number**.
- Now again **save** the Form.



The screenshot shows the 'Tables' list page in ServiceNow. The table below lists various system tables:

Label	Name	Extends table	Extensible	Updated
Account Subscription Entitlement	account_subscription_entitlement	(empty)	false	2026-06-12 14:32:23
Adaptive Authentication Event	adaptive_auth_event	(empty)	false	2026-06-12 14:41:06
Agent Assist Recommendation	agent_assist_recommendation	Application File	false	2026-06-12 14:35:51
MID Server File	agent_file	(empty)	false	2026-06-12 14:22:34
AIG Access Policy	aig_access_policy	Application File	false	2026-06-12 14:01:27
AIG Access Policy Scope Mapping	aig_access_policy_scope_mapping	Application File	false	2026-06-12 14:01:28
AIG Scope Tool Mapping	aig_scope_tool_mapping	Application File	false	2026-06-12 14:01:26
Record Producer Configuration	aka_rp_config	Application File	false	2026-06-12 14:39:11
Search Actions	aka_sl_action	Application File	false	2026-06-12 14:39:11
AI Search ACL Overrides	aka_ad_overrides	Application File	false	2026-06-12 14:03:37
AI Search Active Table Ingestion Tracker	aka_active_table_ingestion_tracker	(empty)	false	2026-06-12 14:03:35
AIx Api Transaction Log	aka_api_transaction_log	(empty)	false	2026-06-12 14:03:29
AI Search Async Genius Result	aka_async_genius_result	(empty)	false	2026-06-12 14:03:27
AI Search Async Request	aka_async_request	(empty)	false	2026-06-12 14:03:40

At the bottom, there is a pagination bar showing '1 to 20 of 6,138'.

- Then Save the Form and Under Related Links Click **Show Form**.
- Now Open the Configure On clicking the Context Menu and then Open the **Form Designer**.
- Then Create Fields as Required.

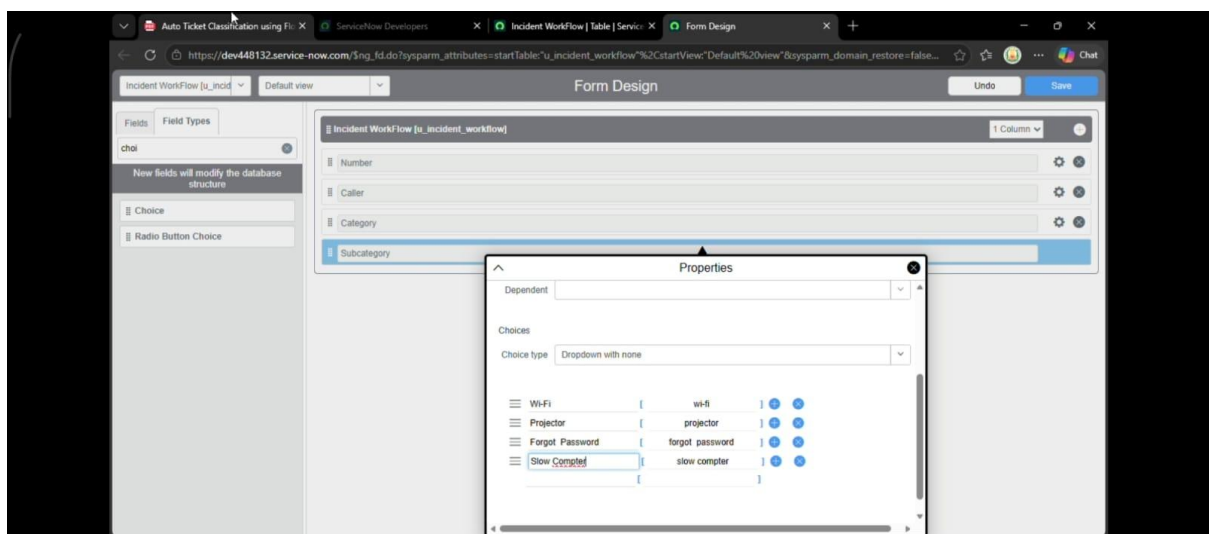
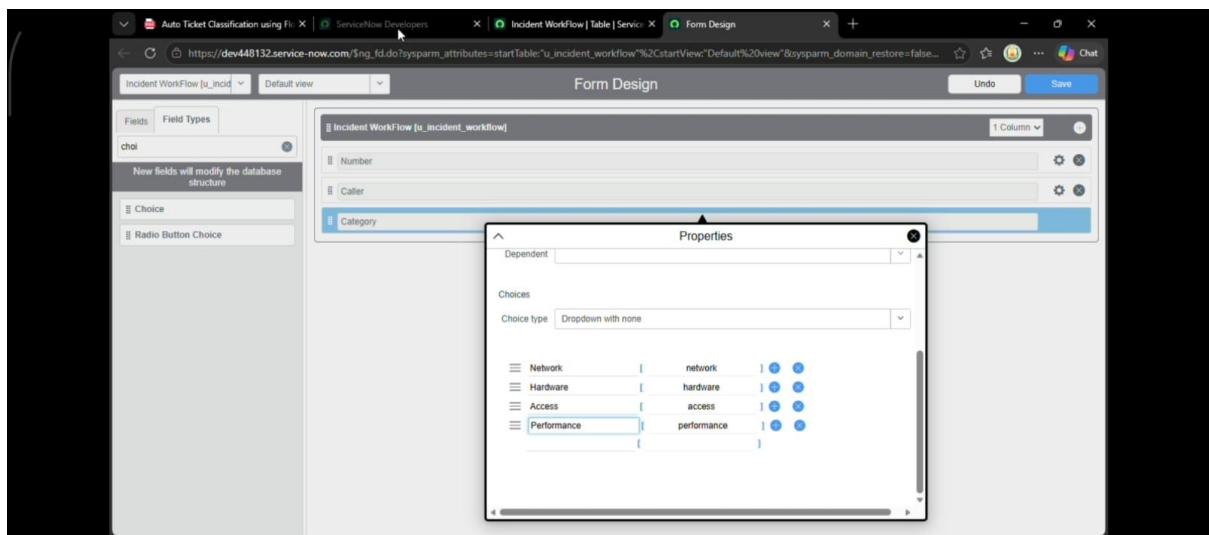
Activity 2: Field Creation and Data Type Configuration

- The objective of this milestone is to create all required fields in the School IT Ticket table with correct data types, choices, and dependencies, ensuring structured and consistent data capture.

Field Label	Type	Reference/Choice
Number	Auto Number	—
Caller	Reference	sys_user
Category	Choice	[Network,Hardware,Access, Performance]
Subcategory	Choice	[Wi-Fi,Projector,Forgot Password,Slow Computer]
Short Description	String	—
Description	String	—

State	Choice	[New,In progress,On hold,Resolved,Closed]
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

- After Creating Fields with their appropriate Type, then add the Choice from the Form design.
- To all the Choice list Fields.
- They are Category, Subcategory and State.



- Same like that, add Choice to the State field.
- Then Click the save button on the top right corner.

- Open the Form and reload it.

Activity 3 : Implementing Dependency Between Category and Subcategory Choice Fields

The objective of this milestone is to implement a **dependency relationship** between the **Category** and **Subcategory** fields so that the available Subcategory options dynamically change based on the selected Category.

Navigation Steps :

- Open the Created Table Form and Right click on the **Subcategory** Field.
- Click on the **Configure Dictionary**.
- Under **Related links** Click on the **Advanced view**,
- Then select the Dependent **Field** Section.
- Enable the checkbox **Use dependent field true**.
- Then select the Field **Category** on the **Dependent on field**.
- Then **save** the Form.

The screenshot displays the ServiceNow 'Incident Workflow' form configuration page. The form is in 'New record' mode. The 'Subcategory' field is selected, and a context menu is open, showing the 'Configure Dictionary' option highlighted. The form includes fields for Number (INC0001001), Caller, Category (set to '-- None --'), Subcategory, Short Description, Description, Assign, and Assigned to. A 'Submit' button is located at the bottom left of the form.

dev310195.service-now.com/sys_dictionary.do?sys_id=998136a6935071081243d5efaba1055&sysparm_view=advanced

Dictionary Entry
Subcategory View: Advanced

Read only option: -- None --

Attributes: edge_encryption_enabled=true

Choice List Specification | Dependent Field | Calculated Value | Default Value

Specify a field whose value is used to determine choice list options for this field. [More Info](#)

Use dependent field: ☒

Dependent on field: Category

Create Choice List | Delete Column | Update

- Now scroll down to the choice section.
- Then add the dependent fields for which option dependent on the category gives the category choice backend name to the subcategory choice values.
- Wi-Fi → Network.
- Projector → Hardware.
- Forgot password → Access.
- Slow computer → Performance.

dev310195.service-now.com/sys_dictionary.do?sys_id=998136a6935071081243d5efaba1055&sysparm_domain=null&sysparm_domain_scope=...

Dictionary Entry
Subcategory View: Advanced

Access Controls | Choices (4) | Attributes (1) | Labels (1)

Label Search

Actions on selected rows... New

Label	Value	Language	Sequence	Inactive	Updated	Dependent value
Forgot Password	forgot password	en	2	false	2026-09-16 02:56:47	Access
Wi-Fi	wi-fi	en	0	false	2026-09-16 02:56:47	Network
Slow Computer	slow computer	en	3	false	2026-09-16 02:56:47	Performance
Projector	projector	en	1	false	2026-09-16 02:56:47	Hardware

Insert a new row...

1 to 4 of 4

- Then save the Form.
- Then it will display each choice to each one separately based on the category subcategory displays.

dev310195.service-now.com/u_incident_workflow.do?sys_id=-1

Incident Workflow New record

Number: INC0001002

Caller:

Category: -- None --

Subcategory: -- None --

Short Description: Hardware, Access, Performance

Description:

State: -- None --

Assigned Group:

Assigned to:

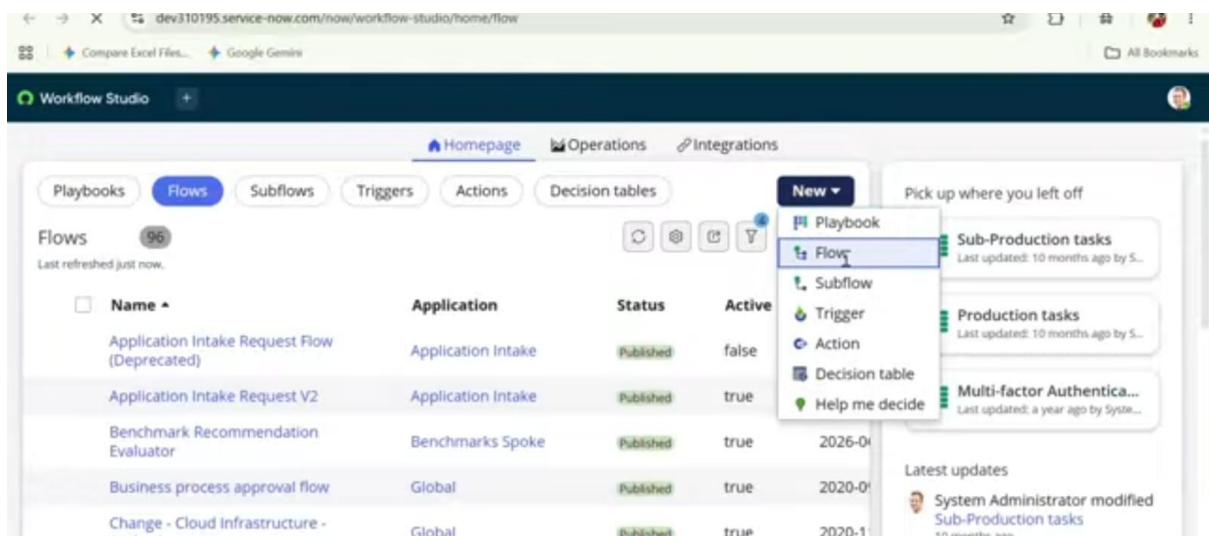
Submit

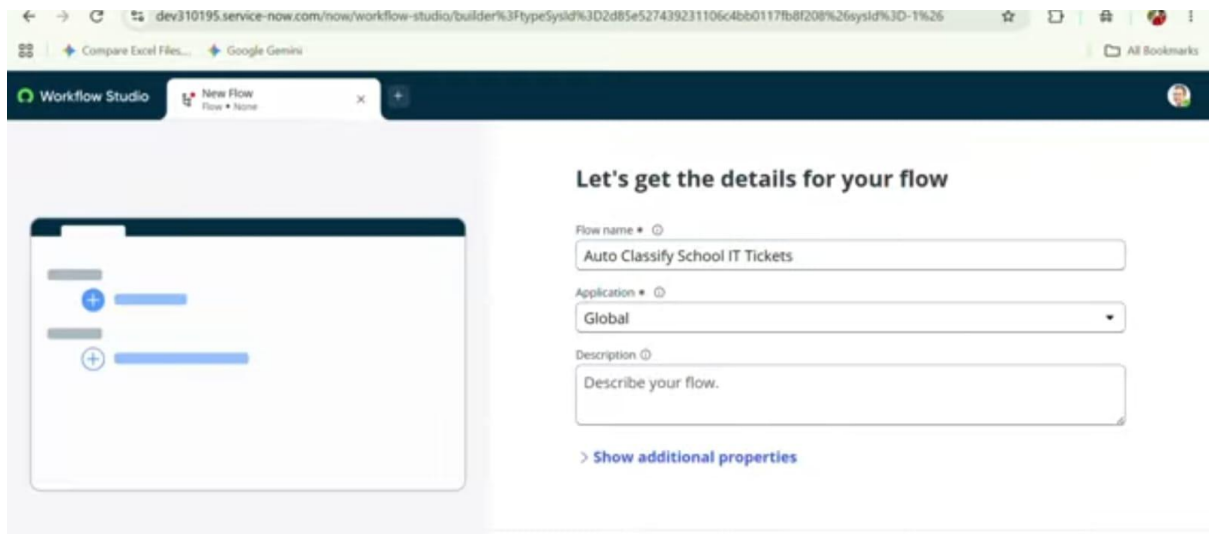
Milestone 3 : Automation using Flow Designer & Email Notification.

When a student creates a new IT support ticket, they only need to enter the Caller Name and Short Description of the issue. The student does not need to manually select the Category or Subcategory. Once the ticket is submitted, a Flow Designer automation is triggered and evaluates the Short Description for predefined keywords such as WiFi, Projector, Password, or Slow. Based on the identified keyword, the flow automatically assigns the appropriate Category and Subcategory to the ticket.

- If the short description contains the word *WiFi*, the flow assigns **Category = Network** and **Subcategory = WiFi**.
- If the description contains *Projector*, the flow assigns **Category = Hardware** and **Subcategory = Projector**.
- If the description includes *Password* or *Login*, the flow assigns **Category = Access** and **Subcategory = Forgot Password**.
- If the description mentions *Slow* or *Hanging*, the flow assigns **Category = Performance** and **Subcategory = Slow Computer**.

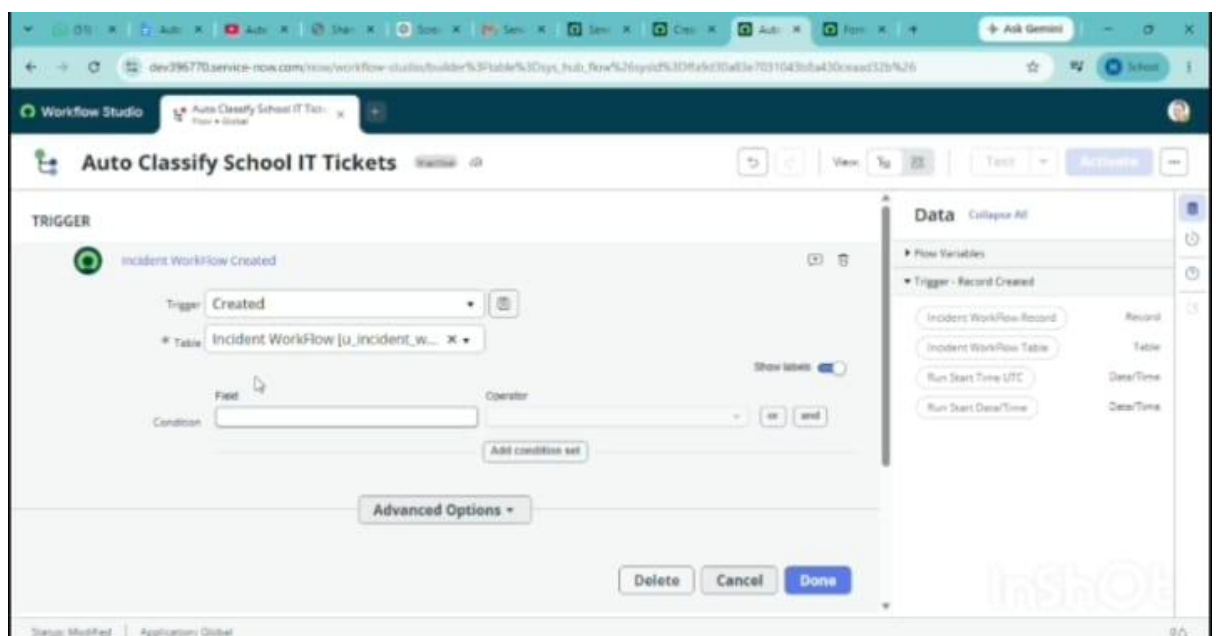
This automation ensures consistent classification, eliminates human error, and significantly reduces the manual effort required by IT staff.





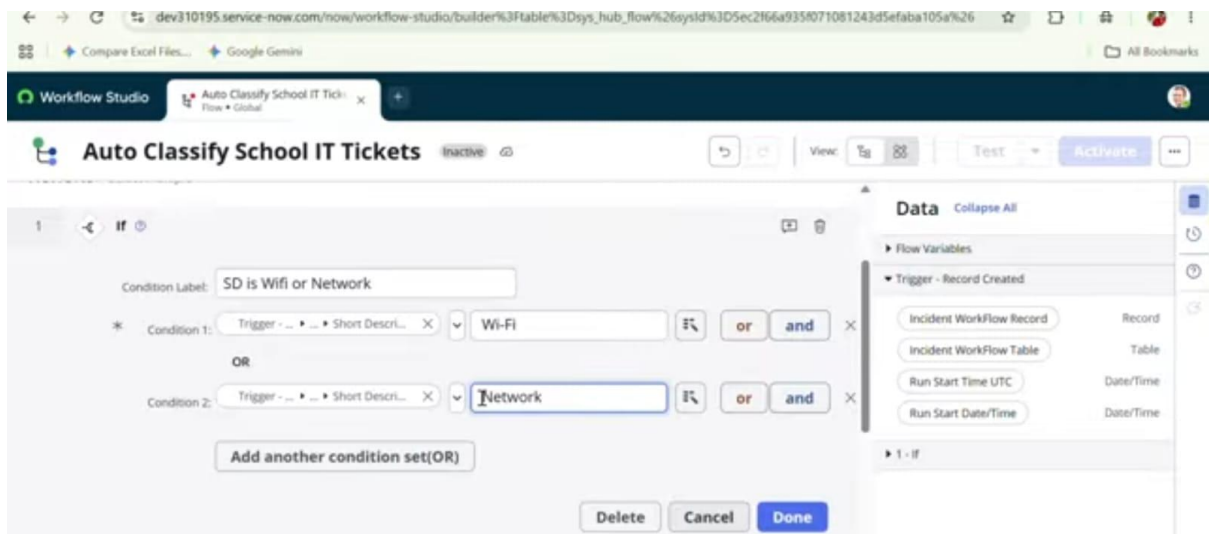
Activity : 2 Trigger Configuration.

- Trigger : **Record Created.**
- Table : **Incident Workflow.**
- Condition : **Category is Empty.**
- Then Click **Done.**
- Now Under **Actions** Select **Flow Logic** Option.

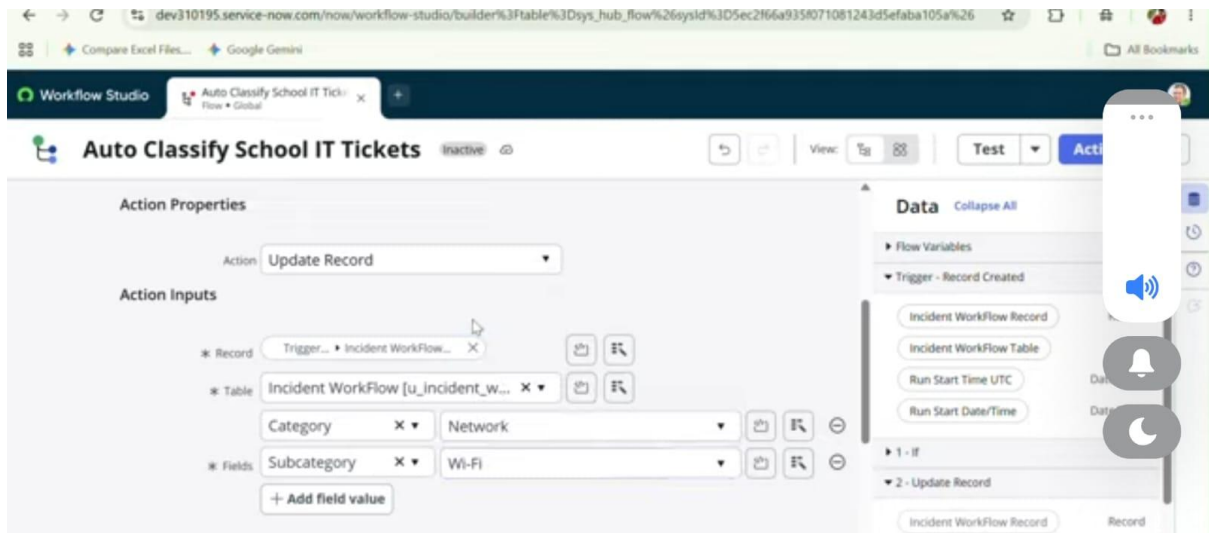


Activity : 3 Wi-Fi Issue Classification.

- Select **If** condition.
- Condition label : SD is **Wifi or Network** (Give as ur wish).
- Condition 1 : Using data pill picker Select **Short description** as Field , and operator **Contains** , Value is **Wi-Fi**.
- Condition 2 : Using data pill picker Select **Short description** as Field, and Operator is **Contains**, Value is **Network**.
- Then Click **Done** Button.

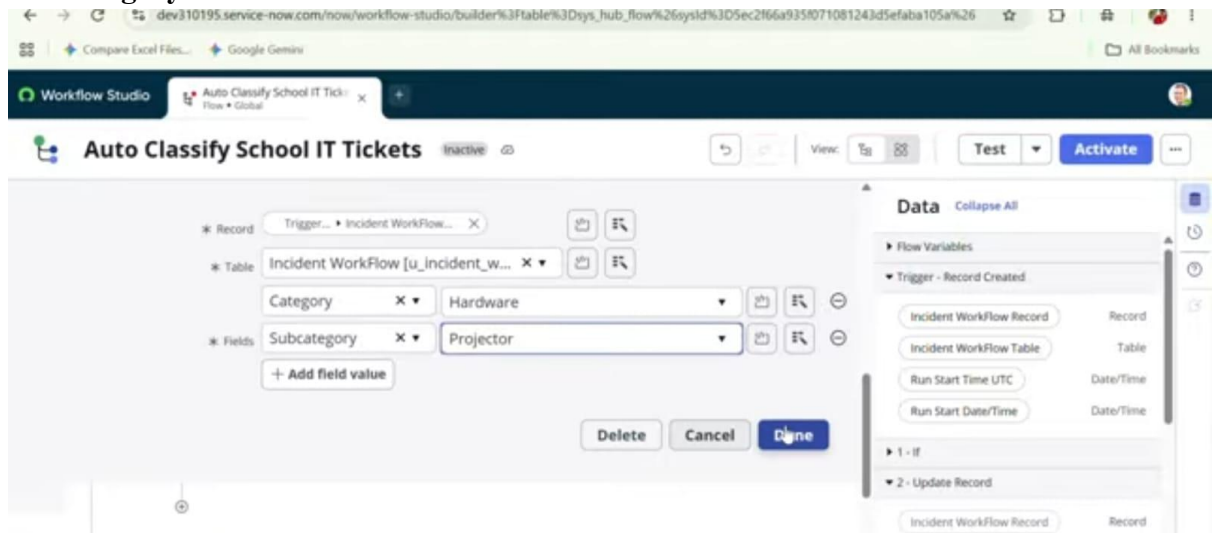


- Now Click **Action to Update record**.
- Action Inputs.
- In Record Field drag and drop data from the right side.
- Table : Auto updates as Incident Workflow.
- Fields Set **Category** = **Network**.
- Set **Subcategory** = **Wi-Fi**.
- Then Click the **Done** button.



Activity : 4 Projector Issue Classification.

- Now Select the **Else If** Condition.
- Condition label : SD is **Projector or Hardware**.
- Condition : **Short description Contains Prajector**.
- Then Click the **Done** button.
- Now Action is an **Update record**.
- Set **Category=Hardware**

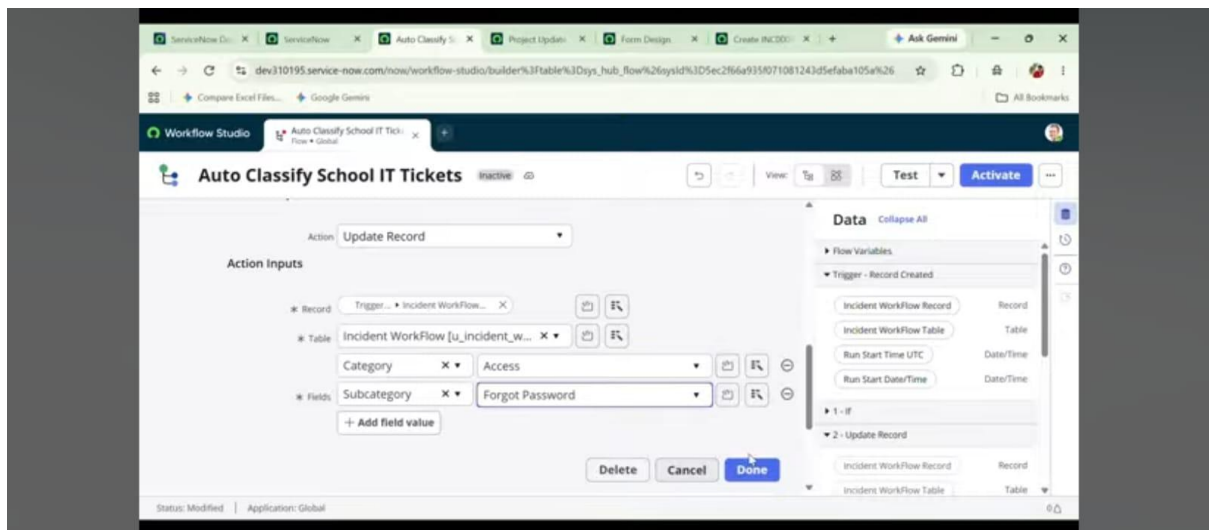


- Set **Subcategory = Projector**.
- Then Click the **Done** button.

Activity : 5 Password Issue Classification.

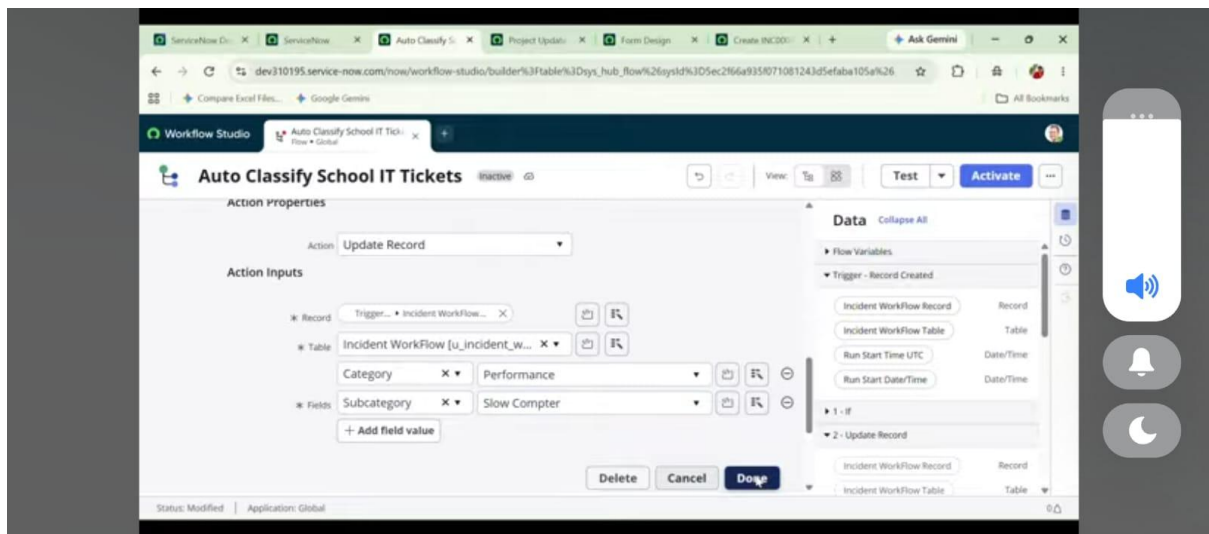
- Now again select **Else If** condition.

- Condition label : SD is **Forgot password**.
- Condition : **Short description contains Forgot password**.
- Then Click the **Done** button.
- Now Action is an **Update Record**.
- Set **Category** = **Access**.
- Set **Subcategory** = **Forgot Password**.
- Then Click the **Done** button.



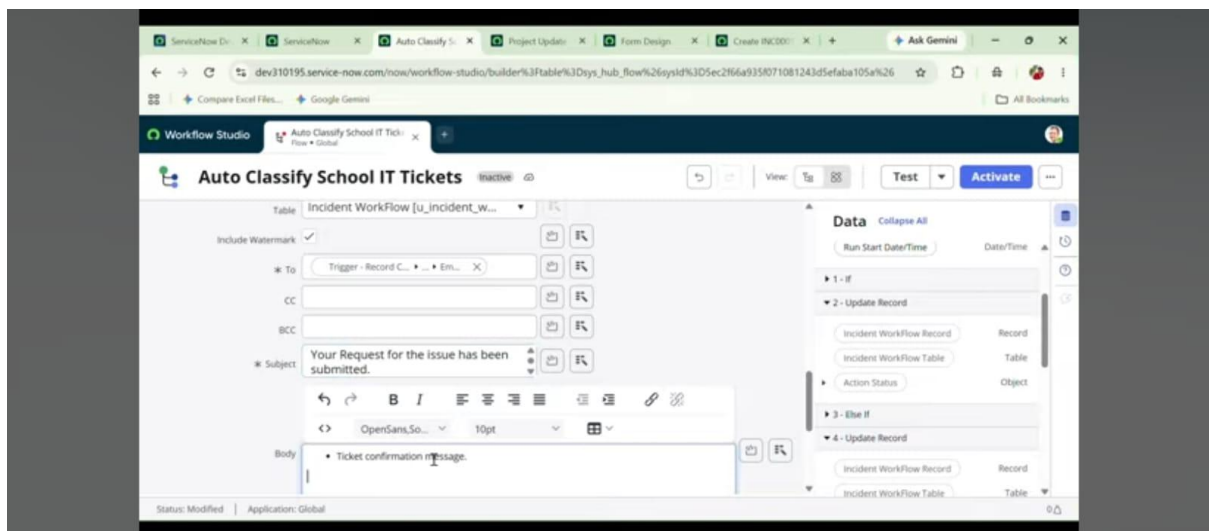
Activity : 6 Slow Computer Classification.

- Now Again Click the **Else If** Condition.
- Condition label : SD is **Performance**.
- Condition : **Short description contains Slow Computer**.
- Then Click the **Done** button.
- Now Action is an **Update Record**.
- Set **Category** = **Performance**.
- Set **Subcategory** = **Slow computer**.
- Then Click the **Done** button.



Activity : 7 Email Action Setup.

- After classification, an email notification is sent using Flow Designer.
- Set Action **Send Email**.
- To : **Caller → Email**.
- Subject : **Your Request for the issue has been submitted.**
- Body : **Ticket confirmation message.**
- Then Click the **Done** button.



- Now Click the **Activate** button from the top right corner.

PHASE 6 : Testing, Validation & Security.

- The objective of Phase 4 is to ensure that the developed solution works as expected, meets all functional requirements, and follows basic security and access control best practices.
- This phase validates the end-to-end automation, confirms data accuracy, and ensures that only authorized users can access or modify ticket data.

Data Validation Checks

- Mandatory fields captured correctly.
- Auto-number generated without duplication.
- Category and Subcategory values stored accurately.
- Reference fields resolve correctly to user and group tables.

Activity : 1 Testing & Validation.

Test Scenario 1

- Enter one Caller Name.
- Short Description: WiFi not working in library
- Then Save the Form.
- And Reload the Form.
- Then it fills auto as :
- Category → **Network**.
- Subcategory → **WiFi**.
- Email sent to caller Email Id.

Activity : 2 Check Email Notification.

- Open All menus → Search Emails.
- Under System logs → Select Emails.
- Search with the Given Subject Name.
- Open that Record and Under Related Links Click Preview mail Option.

Activity : 3 Test Scenario 2

- Enter one Caller Name.

- Short Description: Projector not turning on.
- Then Save the Form.
- And Reload the Form.
- Then it fills auto as :
- Category → **Hardware**.
- Subcategory → **Projector**.
- Then Email sent to caller Mail id.

PHASE 5: Deployment & Final Documentation.

- The objective of Phase 4 is to deploy the developed solution to the target environment and finalize all project documentation.
- This phase ensures that the solution is stable, reusable, and ready for demonstration or production use, along with complete and clear documentation for future reference.

Activity : 1 Make Update set to Complete State.

- Go to All menus → Search For **Local Update Sets**.
- Under **System Update sets** Select → **Local Update sets**.
- Open the **Project Update set**.
- Make the State From **In progress** to **Complete**.
- Then **Save** the Form.
- Under Related links Click **Export to XML** Option.
- Then the Xml Format the Update set data will be **downloaded** to your system.
- We can share through WhatsApp or Mail that downloaded file.
- Then the Created Update set will transform to others in this way.

Conclusion

- The **Auto Ticket Classification using Flow Designer** project successfully delivers a complete, end-to-end automation solution for a school IT helpdesk. The primary objective of reducing manual effort in ticket categorization and improving response efficiency has been fully achieved.
- Through structured requirement analysis, careful backend configuration, and no-code automation using Flow Designer, the system now automatically classifies IT tickets based on the short description provided by students. The implementation of dependent choice fields ensures data accuracy, while automated email notifications improve communication by instantly acknowledging ticket creation to the caller.
- The solution was developed using a phased approach, ensuring clarity, stability, and ease of maintenance. Comprehensive testing and security validation confirmed that the system functions reliably, maintains data integrity, and enforces appropriate access controls.
- Overall, this project demonstrates how ServiceNow can be effectively used to build scalable, maintainable, and user-friendly automation solutions without relying on complex scripting or machine learning. The design is well-suited for small to medium environments such as educational institutions and can be easily extended in the future to include advanced features like assignment automation, SLA tracking, or predictive intelligence.